

Assignment 2
Spring 25
CSE 350

Instruction:

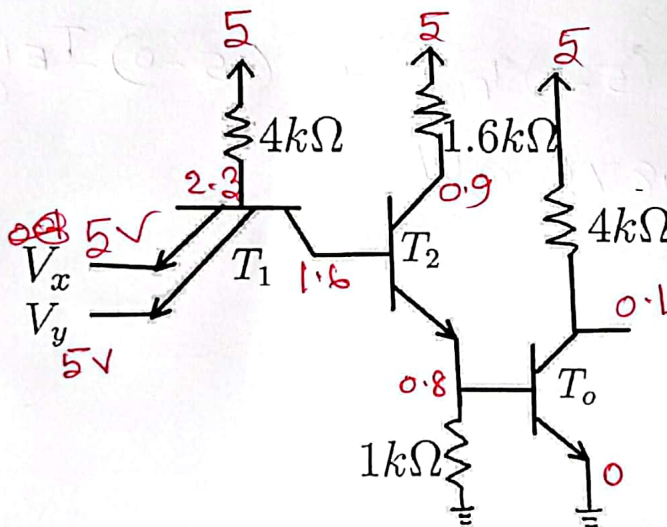
- ✓ The solution must be in a hand-written format
- ✓ PDF file of the solution needs to be submitted through Google form

Question 1

Marks [10]

Identify the Logic gate given below and write its logic expression. For the given TTL circuit find the power dissipation for all the cases.

For circuit given below, $V_{CE(Sat)} = 0.1 V$, $V_{BE(Sat)} = 0.8 V$, $\beta_F = 30$, Logic low = $0.1 V$, Logic high = $5 V$, $V_{BC(RA)} = 0.7 V$, $\beta_{RA} = 0.1$



This is TTL NAND gate.

Case: 00

$$I_{01} = \frac{5 - 2.3}{4k}$$

$$P = (5 - 0.1) \times \frac{5 - 2.3}{4k}$$

$$= 3.308 \text{ mW}$$

Case: 01, $P = 3.308 \text{ mW}$

Case: 10: $P = 3.308 \text{ mW}$

Case: 11

$T_1 \rightarrow$ Reverse active mode

$$I_{B1} = \frac{5-2.3}{4k} = 0.675 \text{ mA}$$

$$I_{C2} = \frac{5-0.9}{1.6k} = 2.5625 \text{ mA}$$

$$I_{C0} = \frac{5-0.1}{4k} = 1.225 \text{ mA}$$

$$I_{E1x} = I_{E1y} = 0.1 I_{B1} = 0.0675 \text{ mA}$$

$$\begin{aligned} P &= (5-0) I_{B1} + I_{C2} (5-0) + (5-0) I_{C0} \\ &\quad + (5-0) I_{E1x} + (5-0) I_{E1y} \\ &= 22.9875 \text{ mW} \end{aligned}$$